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CUSTOMIZATION OF LLM-SPECIFIC MODULES WITHING CHATGPT

BACHELOR THESIS

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Declaration

I hereby declare that I am the sole author of the bachelor thesis “Customization of LLM-specific modules withing ChatGPT”. I duly marked all of the quotations and all used literature and sources are stated in the attached list of references.

In Prague on

Signature

Abstrakt

Cílem dané bakalářské práce je vyvinuti dvou odlišných speciálních verzí ChatGPT určené pro konkrétní moduly a vyhodnotit, zda tyto přizpůsobené verze fungují lépe a jsou snazší na používání ve srovnání se standardním ChatGPT. Pro hodnocení a srovnání budou pozváni experti z příslušných oborů, kteří tyto velké jazykové modely otestují a poskytnou své názory.

Motivací pro tuto práci je rostoucí zájem a možnosti pro vytváření přizpůsobených verzí ChatGPT. Klíčovými aspekty jsou použitelnost, rozdíly oproti standardnímu modelu a snadnost používání. Kromě toho je zde osobní zájem autora zjistit, zda zůstanou vytvořené přizpůsobené modely užitečné i po dokončení této práce a zda nabídnou jakékoli budoucí aplikace.

Tato práce je rozdělena do dvou hlavních částí: teoretické a praktické.

Teoretická část poskytuje komplexní přehled a úvod do velkých jazykových modelů, popisuje specifika generativních předtrénovaných transformátorů. Zabývá se zásadním konceptem promptů, diskutuje, jak je mohou uživatelé efektivně využívat. Dále tato část popisuje složitosti jemného ladění a zásady RAG v kontextu těchto modelů.

Praktická část práce popisuje vývoj dvou odlišných speciálních verzí ChatGPT. První verze je navržena tak, aby pomáhala s tvorbou životopisů, zatímco druhá je přizpůsobena pro pomoc uživatelům v přípravě na test znalosti japonského jazyka (JLPT) a také učitelům s plánováním hodin a školními úkoly. Tato část nejen podrobně popisuje proces vytváření, ale také zkoumá konkrétní funkcionality a aplikace každé přizpůsobené verze.

Ve závěru bakalářské práce byly shrnuty výsledky hodnocení přizpůsobených modelů ChatGPT, včetně celkového hodnocení každé verze na základě jejich výkonu. Závěr také nastínil potenciální aplikace pro tyto specializované GPTs, navrhuje způsoby, jak by mohly být efektivně začleněny do příslušných oblastí za účelem zlepšení úkolově specifických funkcionalit a zkušeností uživatelů.

Klíčová slova

LLM, ChatGPT, AI, GPTs, Přizpůsobitelnost, Customizace

Abstract

The goal of the thesis is to develop two distinct custom versions of ChatGPT tailored for specific modules from scratch, and to evaluate whether these customized versions perform better and are easier to use compared to the standard ChatGPT. For the evaluation and comparison, experts in relevant fields will be invited to test these large language models and provide their insights.

The motivation for the thesis stems from the growing interest in and possibilities for creating customized versions of ChatGPT. Key considerations include usability, differences from the standard version, and ease of use. Additionally, there is a personal interest of the author in determining whether the custom models created will remain useful beyond the completion of this thesis and if they will present any future applications.

Thesis is divided into two main sections: theoretical and practical.

The theoretical part provides a comprehensive overview and introduction to large language models, describing the specifics of Generative Pre-trained Transformers. It covers the essential concept of prompts, discussing how they can be effectively utilized by users. Additionally, this section describes the complexities of fine-tuning and the principles of RAG in the context of these models.

The practical part of the thesis describes the development of two distinct custom versions of ChatGPT. The first version is designed to assist with creating resumes, while the second is tailored to aid users in preparing for the Japanese Language Proficiency Test (JLPT), as well as helping teachers with lesson planning and school-related tasks. This section not only details the creation process but also examines the specific functionalities and applications of each customized version by the experts.

In the conclusion of the thesis, the evaluation results of the customized ChatGPT were summarized. The conclusion also outlined potential applications for these specialized GPTs, suggesting ways they could be effectively integrated into relevant fields to enhance task-specific functionalities and user experiences.

Keywords

LLM, ChatGPT, AI, GPTs, Adaptability, Customization

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List of Abbreviations

LLM	large language models
RAG	retrieval-augmented generation
GPT	generative pre-trained transformer
JLPT	Japanese Language Proficiency Test
AI	Artificial Intelligence

Introduction

The trend of using ChatGPT emerged and began to actively develop relatively recently. The first version of this chatbot appeared on October 30, 2022 (OpenAI, 2021). The primary audience using ChatGPT is traditionally considered to be people of all ages, with most users of age between 25-34 years old (similarweb.com). This demographic finds the program's capabilities particularly useful for various purposes.

For individuals in the main age group, ChatGPT serves as a useful tool for both, professional and personal tasks. Professionals often use ChatGPT to draft reports, emails, and other business correspondence efficiently. They may also leverage its capabilities to brainstorm ideas, solve complex problems, or generate content for all kinds of campaigns. Moreover, individuals, who are pursuing higher education or engaged in continuous learning find ChatGPT valuable for academic purposes. It can assist in research tasks, help with writing academic papers, and provide explanations or insights on complex topics.

“Despite the high popularity and widespread use of chatbots, they often struggle to perform most tasks efficiently and effectively. This can be attributed to various factors, including limited natural language processing (NLP) skills, insufficient training on a broad range of data, or restricted capabilities in user interaction. For example, language models are prone to saying things that are false, a problem called hallucination. Language models are trained to generate text that is predictable and coherent, but the training algorithms we have seen so far don’t have any way to enforce that the text that is generated is correct or true. This causes enormous problems for any application where the facts matter!” (Daniel Jurafsky, 2024).

The problem lies in the fact that ChatGPT operate with a limited amount of data and rules upon which it can generate responses. This can lead to situations where a chatbot fails to accurately interpret a user's question, provide irrelevant response or generate output which will contain mistakes.

One of the key factors influencing the quality of a chatbot's response is the level of development of its natural language processing algorithms. In some cases, these algorithms may fail to recognize nuances in the question or overlook the context of the conversation, resulting in incorrect, incomplete or offensive answers. For example, HSBC company launched its chatbot, which was aimed to help customers with general questions about bank services. Even though it was able to solve simple tasks, more complicated questions and nuanced requests started to be a struggle (Michelle Parayil, 2022). Another problem happened in the year 2020, when a Parisian healthcare facility tested GPT-3 to give made-up patient medical advice. During the test, a simulated patient expressed severe distress and suicidal thoughts, stating they felt very bad and wanted to end their life. GPT-3's response was alarmingly inappropriate, initially indicating it could assist the patient with their desire for self-harm (Ryan Daws, 2020).

Motivation for the thesis

The primary motivation for this thesis is to expand capabilities in creating and manipulating GPTs, thereby offering the opportunity to develop personalized tools for specific needs. The thesis will focus on developing two specialized GPTs. The first model is tailored for creating resumes and CVs, aiming to help with the job application process—a task that the author personally found challenging while applying to multiple positions simultaneously. This GPTs will not only save time for job seekers but also assist in overcoming common challenges faced during job applications. Additionally, it will explore the capability to distinguish between documents generated by AI and those written by humans.

The second model is designed for helping with the preparation for the Japanese Language Proficiency Test (JLPT). This motivation stems from a personal request from the author's language teacher, as well as the author's family residing in Japan and personal language learning journey. The goal for this GPTs is to understand how custom AI tool can enhance the learning experience for the language test candidates, specifically examining how it compares to conventional study methods and standard versions of ChatGPT. This GPTs aims to provide tailored support during the preparation period, adapting to the user's learning progress and specific exam preparation needs.

Objectives of the thesis

This thesis is dedicated to the development and evaluation of customized versions of ChatGPT, each designed for specific applications. The performance of these models will be examined in terms of accuracy, efficiency, and response quality. By comparing these specialized GPTs to the standard ChatGPT, particularly focusing on their ability to interact and respond to user queries, the study aims to discover if the customized versions deliver more relevant and effective responses. Key objectives include:

- Creating two custom versions of ChatGPT targeted at different fields to assess their overall performance.
- Evaluating the performance of these GPTs with the assistance of experts in relevant fields.
- Analyzing feedback and scores from these experts to determine if the customized versions outperform the standard model.
- Exploring potential future applications for these customized GPTs and assessing their long-term utility.

The outcome will help determine the necessity and value of customized ChatGPT for specific tasks and their potential for broader applications.

Research methods

Research methods of this thesis include:

1. For evaluating the quality, effectiveness, and usefulness of the GPTs, the author set up interviews with experts from relevant fields. The feedback was collected through two primary methods: face-to-face communications, which included both in-person and online meetings requested by some professionals who could not test the GPTs independently, and email communications. The second option was particularly used for reviewing the JLPT GPTs, due to significant time differences. All candidates were provided with the necessary evaluation criteria and questions, and they returned their feedback to the author via email.
2. For the creation of the GPTs, the author utilized the "My GPTs" section on the chat.openai.com website. This platform is specifically designed for experimenting with, creating, and testing custom versions of ChatGPT, providing users with the tools and interface necessary to tailor GPTs to specific needs and applications.

1 Evolution of Large Language Models

1.1 Introduction to Large Language Models

Large Language models (LLMs) – advanced type of language model that is trained using deep learning techniques on massive amounts of text data. These models are capable of generating human-like text and performing various natural language processing tasks (Suvojit Hore, 2024).

Large Language Models can be known for their size, with hundreds of millions or even billions of parameters, allowing them to capture complex patterns in human language. These models predict the next word in a sequence, learning syntax, context, and semantics from text. This capability makes them very effective at understanding and generating detailed and nuanced text, which is useful for a wide range of natural language processing (NLP) tasks (Humza Naveed, 2024)

Training a Large Language Model involves feeding it a vast dataset that includes a variety of text sources, such as books, websites, and articles, spanning multiple languages and fields. This stage, known as pre-training, usually requires powerful computing resources and can last weeks or months. After this, LLMs go through a process called fine-tuning. This is a more focused training phase where they are taught using smaller, specialized datasets to enhance their performance on specific tasks like sentiment analysis, summarization, or question answering (Jacob Devlin, 2018).

1.1.1 LLM History

The foundation of Large Language Models (LLMs) can be traced back to the early days of neural networks and information processing systems in the 1950s. During this time, researchers were focused on developing computer systems capable of processing natural language. One of the early challenges was creating systems that could automatically translate phrases and texts from one language to another, such as from Russian to English.

One of the earliest examples of a computer program utilizing a basic form of language model was "Eliza," developed in 1966. Eliza's main function was to provide "psychological advice" to users based on the prompts given to it. While rudimentary by today's standards, Eliza represented an early attempt to simulate human-like interaction through text-based communication.

OpenAI has released five versions of its Generative Pre-trained Transformers (GPT) series, namely GPT-1, GPT-2, GPT-3, GPT-3.5, and the latest addition, GPT-4. The initial version, GPT-1, was introduced in 2018, with subsequent releases aimed at enhancing various aspects of natural language processing tasks such as text generation and language understanding.

One of the particular significances are the latest iterations, GPT-3.5 and GPT-4.0, which have seen remarkable improvements in the capabilities of Large Language Models (LLMs). In the fourth version, in particular, LLMs have demonstrated remarkable advancements. For instance, they now possess the ability to interpret images, analyze CSV files, generate more precise code, and even craft website content, among other functionalities (OpenAI, 2023)

Fast forward to today, and people have access to much more sophisticated chatbots powered by advanced LLMs. Unlike their predecessors, modern language models are capable of performing a wide range of tasks with remarkable accuracy and efficiency. For example, they can classify text based on sentiment, distinguishing between negative, positive, and neutral moods. This capability allows them to tailor responses according to the emotional context of the conversation.

Moreover, modern LLMs can also consider ethical and legislative conditions when generating responses. They can analyze the prompt given by the user and determine whether the information provided aligns with ethical guidelines or legal requirements. This ability to contextualize responses contributes to the overall reliability and usefulness of modern chatbots.

In essence, the evolution of language models from the basic systems of the past to the sophisticated LLMs of today represents a significant advancement in the field of natural language processing. These models have revolutionized the way we interact with computers and have opened new possibilities for automation and efficiency in various domains.

1.1.2 Types of LLMs

The field of Large Language Models has undergone fast expansion and notable success over recent years. With the development of numerous LLMs tailored for various purposes, it's essential to examine selected models to gain a deeper understanding of the broad interest in and practical applications of these technologies.

- BERT - Bidirectional Encoder Representations from Transformers – this model was created by Google, and it stands out, mostly, because of its ability to understand the context of words by examining the words that comes before and after them. This approach makes BERT particularly effective for tasks requiring a deep understanding of language context, such as sentiment analysis and question answering. (Zecong Wang, 2023)
- T5 - Text-To-Text Transfer Transformer – developed by Google, the T5 model reframes all text-based language tasks into a unified text-to-text format. This approach involves converting one type of text into another, making T5 highly versatile and applicable to a wide range of tasks across different domains. (Colin Raffel, 2020)
- RoBERTa - Robustly Optimized BERT Approach – based on the BERT architecture, RoBERTa makes significant modifications to key hyperparameters. It removes the next-sentence pretraining objective and trains with much larger mini-batches and higher learning rates. These

changes enhance RoBERTa's performance, making it more effective at understanding and processing language tasks. (Yinhan Liu, 2019)

- XLNet - This model combines the strengths of autoregressive language modelling and autoencoding. It captures bidirectional context through a permutation-based training method, which differs from traditional sequential approaches. This technique allows it to effectively process and predict language in a way that understands the interdependencies of words in sentences more holistically. (Zhilin Yang, 2020)

1.1.3 Neural Network Architecture and Deep Learning within LLMs

Deep Learning - machine learning approach that allows computers to learn from experience and understand the world by representing knowledge in terms of a hierarchy of concepts. In deep learning, each concept is defined in relation to simpler concepts, and knowledge is gathered from experience rather than being formally specified by human operators (Ian Goodfellow, 2015).

Large Language models are built on the neural network architectures, to be precise, transformer architecture. It allows them to proceed and analyze different language patterns and relationships between words, phrases and sentences within large-scale datasets (Ashish Vaswani, 2023).

1.1.4 Attention mechanisms

Attention mechanisms itself improve deep learning models by selecting the most relevant information in the given dataset or question. They prioritize the most important details in order to enhance overall score of model performance.

Simply said, it works just like human brain. If someone gives a person a picture of seven dogs and one cat, and asks where the cat is, one's brain will be aimed to define only one thing "a cat", ignoring remaining information and objects.

Transformer architecture is based on two attention mechanisms: self-attention and multi-head attention.

Self-attention mechanism – sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations. (Ashish Vaswani, 2023)

Multi-head mechanism – allows the model to jointly attend to information from different representation subspaces at different positions. With a single attention head, averaging inhibits this. (Ashish Vaswani, 2023)

2 Overview of LLMs

2.1 Introduction to Prompts

Prompt is the input, that the user has provided to the AI model. This input serves as an instruction for the model to predict and generate the desired response to the user.

In the pictures below, there is an example of prompts and one of the possible answers on it.

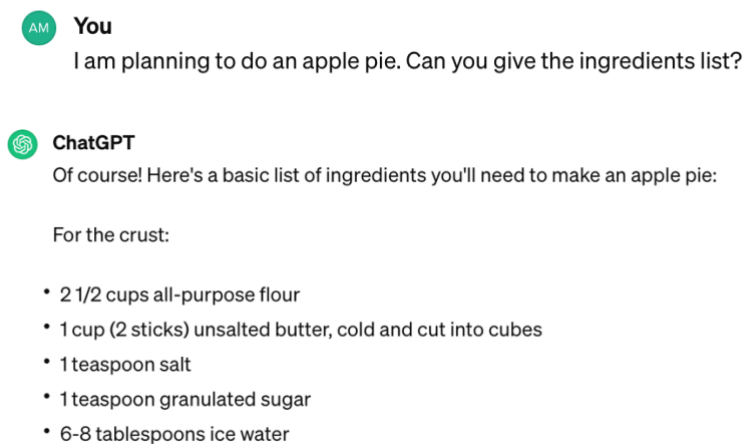


Figure 1 prompt example (source: author)

In the picture above is a perfect example of one of the simplest prompts, that could be given to the ChatGPT and the answer will be more than enough for the everyday users. Even if there are slight errors in the input, they are unlikely to significantly alter the overall response. However, if needed a more difficult task, for example, analysing extensive datasets, performing tough calculations or create specific picture, then every aspect of the question becomes crucial. Factors such as the length of the prompt, the choice of words, and the level of detail provided significantly influence the accuracy and relevance of responses generated by LLMs. Therefore, it becomes necessary to use specialized rules and techniques to optimize the generation process and receive precise answers customized to the specific requirements of the task.

2.2 Type of prompts used in GPTs

Before considering the techniques and strategies of making prompts, there is a need to define the type of ones.

Direct prompting (Zero-shot) – the most basic form of prompts. It provides fundamental and the most important information and instructions but does not contain any practical examples. As a result, the models cannot classify text such as positive or negative without explicit guidance. However, given that the model has

been trained on data containing both positive and negative segments, it can still comprehend the context of the prompt provided. (Takeshi Kojima, 2022)

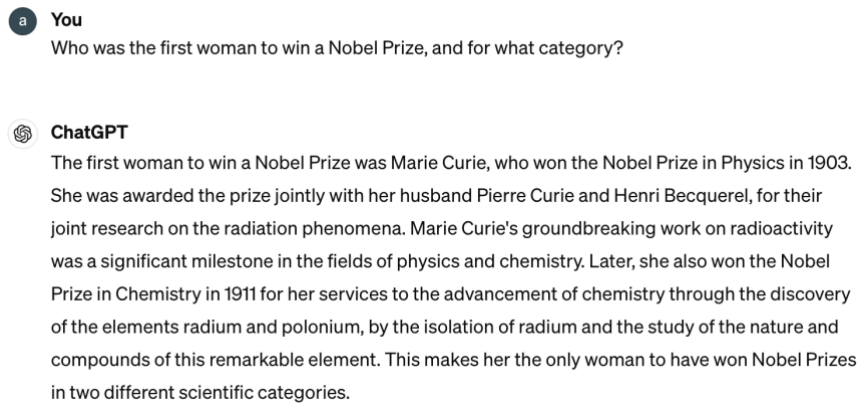


Figure 2 zero-shot prompt (source: author)

Prompt with examples – this type of prompts can be divided on the 3 sections: one-, few- and multi-shot. The only difference between them is the number of examples that would be given to the model, so it can classify the text and give the proper answer for the question.

It means, that in one-shot or few-shot, there will be one to four clear example for the model.

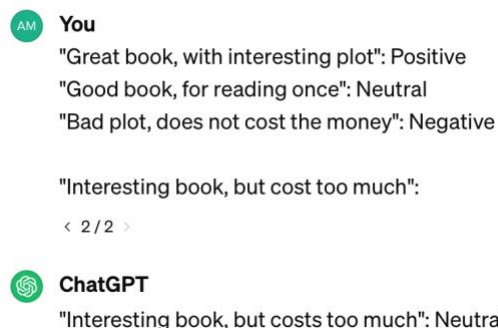
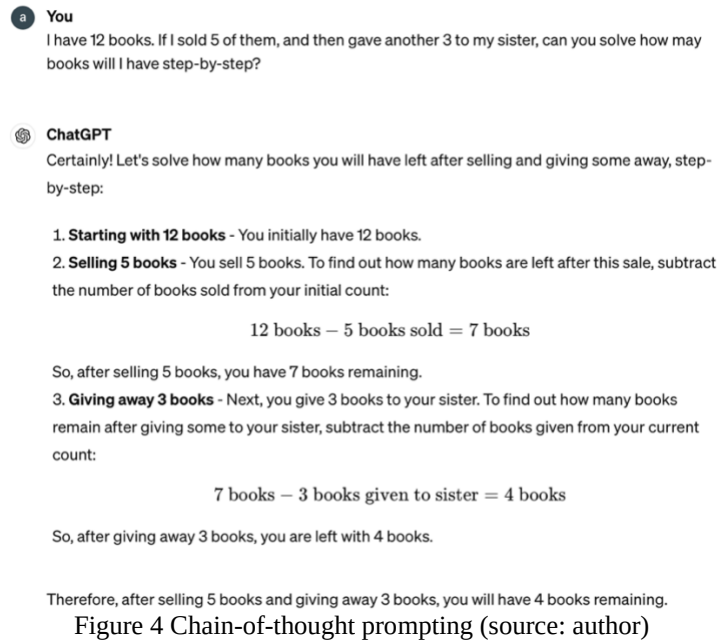


Figure 3 Prompt with examples (source: author)

Chain-of-thought prompting (CoT) – this way of prompting basically means, that LLM will need to explain the reason of the generated and given answer step-by-step. This prompting enables models to solve more complex and large problems through intermediate steps with explanation. (Wei et al. 2022)



2.3 Prompt techniques and strategies

As was said earlier, right strategy and format of the prompts plays huge role in generating answers. The more consent and pure the prompt will be, the clearer and more relatable will be the answer. Even though, there are no rules or restrictions in formulating and designing text, users may still use some simple rules that can affect the model's response.

2.3.1 Give instructions

The most basic and effective way to customize model behaviour for generating answer. Users must be confident that their input is clear and concise, leading to wanted outputs. When assigning tasks to the model, it's crucial to define questions in an understandable manner, providing clear guidance for generating accurate answers.

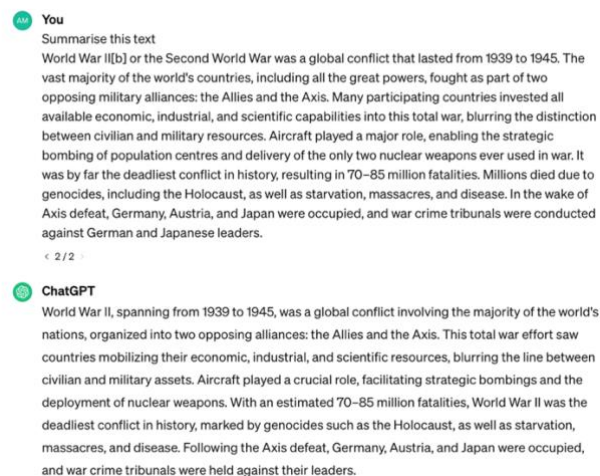


Figure 5 Clear output example (source: author)

Moreover, users have the option to specify any constraints that might influence the generated answer. In the picture above, the text was summarized by the GPT, but if user

want just one sentence to be written as a sum up, then it can be easily done, by specifying the condition.

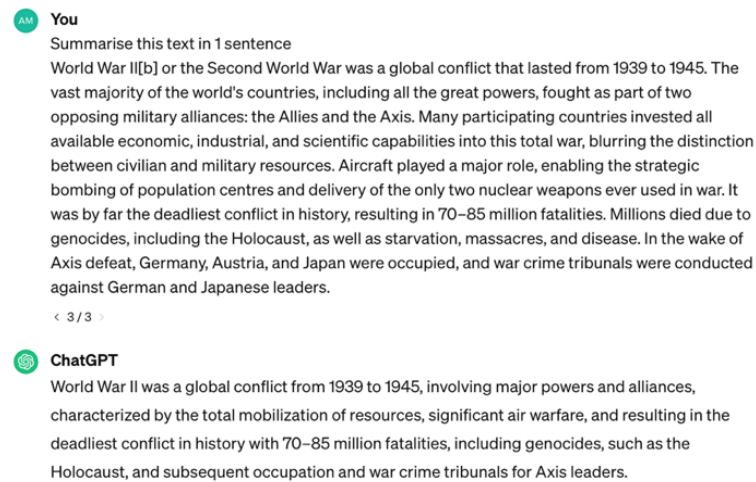


Figure 6 Format of response (source: author)

Another key points of instructions might be the format of response. User can specify, if the answer must be given as a table, bullet point, underlined, keywords marked with bold text and so on. For example, ChatGPT can also be asked to make a 2x2 table with key dates and what events had happened.

2.3.2 Include examples in the prompt

As was mentioned earlier, it's beneficial to include examples of answers in the question so that the model can identify and understand what the desired response should look like. Therefore, instead of relying only on zero-shot prompts, users may opt to use few-shot prompts when expecting related output.

Additionally, it is necessary to mention, that, while giving the few-shots prompts examples, user should avoid changing the format and pattern of the prompts. Otherwise, model can start overfit or “overthink” the answer, which may lead to undesired output.

2.3.3 Add Information

Rather than making assumptions and rewriting questions for models, users can enhance the quality of output by providing additional information and context about the situation. This approach allows models to generate more efficient and relevant responses in less time, as they have a clearer understanding of the task.

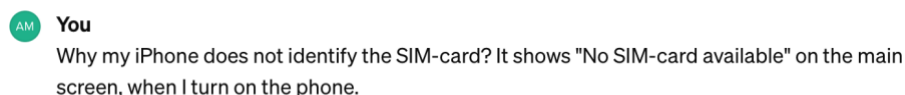


Figure 7 Prompt with additional information (source: author)

As can be seen in the picture above, the additional context can be identified by the words “it shows” and “iPhone”, which give the model specific details for generating output.

2.4 Challenges and consideration

While AI and GPTs have experienced tremendous growth, several challenges persist in their usage, as indicated by statistical data and research findings.

In the world of AI, when an it gives false or misleading information, experts called it "hallucinating." For ChatGPT, this means it can sometimes provide incorrect details, but presents it as it was true, which can be tricky because it's hard to tell when the information is wrong. Users often rely on ChatGPT's answers, but if they trust it too much without checking, it can lead to mistakes or spread wrong information. ChatGPT uses lots of data from the internet, including more reliable scientific sources. But because it mixes all this information, it's challenging to pinpoint when it's making things up (Łukasz Białozor, 2023). The false information might be hidden among true facts, making it hard for users to notice. This is why it is important to be extra careful when using ChatGPT, especially for important decisions or sensitive topics.

Next problem is limited knowledge and databases of GPTs. Even with the fact, that ChatGPT Plus includes fresh information up to April 2023 (seo.ai, 2023), users, that don not pay for the subscription, may notice, that when they will try to search some completely new information, or something that has happened in the past few months, GPTs will more likely start its answer with the warning of not having the actual information about the situation, but then will proceed to give an approximate answer based on the facts that it already knows.

Furthermore, concerns regarding data privacy and security have emerged, with instances of GPTs being exploited to create malware, information stealers, and potentially hazardous code. This issue has escalated with the release of advanced versions like ChatGPT-4, which boast significantly higher success rates in executing code, raising cybersecurity risks (Lauren Laws, 2023).

Next issues can be described as a trust level. Even though, it may seem that peoples' trust to the GPTs could be low, the reality shows the controversial opinions on this question. For example, in the UK the sales sector shows almost 100% lack of trust to the GPTs (Techerati, 2023). However, while talking about human-machine interactions, it said, that there is a risk of over-trusting AI systems, when mentioning something that people do not have many knowledges about. It might appear, because humans' desire to escape hard challenges and decisions, and put responsibility on something else, in this case, artificial intelligence (Harris, 2021).

2.5 Fine-tune Problematics

Fine-tuning in the context of Large Language Models refers to the process of adjusting a pre-trained model to improve its ability to follow specific instructions and generate appropriate, safe responses. This process enhances the model's performance on tasks it was not specifically trained for initially, improves its ability to learn from minimal data, and increases its effectiveness across different tasks. Fine-tuning typically requires only a small additional effort compared to the initial extensive pre-training, making it an efficient way to tailor LLMs to align with desired outcomes and user intents more closely. (Humza Naveed, 2023)

2.5.1 Fine-tuning usage

Fine-tuning is a critical aspect of deploying Large Language Models, enabling users to customize models like GPTs for specific tasks. This process enhances the model's accuracy, relevance, and overall performance by adjusting it to better align with objectives or requirements. (Medium, 2023).

The key benefits of using fine-tuning practice for the models, according to Lev Craig (2024), may include:

1. Utilizing pre-trained models for fine-tuning is cost-effective and efficient. By building on what the model already knows, fine-tuning speeds up learning and delivers quicker results, saving resources.
2. Better results for the small cases. Fine-tuning is especially beneficial for tasks in specialized fields or with small datasets. Customizing the model to meet the specific needs of these tasks improves its performance, leading to more accurate and relevant results. (Craig, 2024)

2.5.2 Requirements for using fine-tuning within GPTs

The very first step for users, after determining the necessity of using fine-tuning, will be preparing the dataset for training the model. User should create dataset with several examples, that must be diverse, but at the same time similar to the conversation that user will have with the model. One of the important moments, at least one of the examples must be aimed at the cases where model will not behave the way it should. The estimated minimum of examples is ten, but the best choice would be the number between fifty and one hundred, so the model can show improvements during fine-tuning.

After preparing the data, it needs to be loaded and possibly modified for the pre-trained model to suit the task—for example, adjusting it to classify sentiments. User proceed to train or fine-tune the model on dataset, allowing it to learn from the provided examples and fine-tuning its parameters to minimize errors. Validation set can be used to monitor the model's performance to avoid overfitting.

Finally, model can be tested on a separate dataset to ensure it generalizes well to new data. Continuously refining the process by adjusting parameters, expanding

the dataset, or tweaking the model architecture, enhances the model's accuracy and effectiveness for specific application. (OpenAI, 2024)

2.6 Retrieval-augmented generation

RAG (Retrieval-augmented generation) is an artificial intelligence framework designed to enhance the quality of content generated by large language models (LLMs). Integrating the RAG framework into LLMs offers numerous advantages. Firstly, it ensures that models can tap into the latest and most relevant information, facts, articles, and sources available. This access to up-to-date data enhances the quality and accuracy of the generated content. Additionally, users are granted access to the sources utilized by the model, fostering transparency and trust in the output. Moreover, RAG prioritizes information that is trustworthy and accurate, thereby minimizing the likelihood of misleading or erroneous content (Martineau, 2023).

2.6.1 RAG process

While using RAG, users enhance a language model's ability to respond by adding an information retrieval component. This component grabs relevant information from external sources based on what the user asks. Both the user's question and the gathered information are then given to the LLM. With this extra knowledge alongside its existing training, the LLM can generate responses that are more informed and fitting to the context (Amazon Web Services, 2024).

As a first step user will provide the question prompt to the model. After that, RAG model will retrieve needed information from its relevant documents and knowledge sources, such as databases. This process aimed to gather all the information related to the given prompt: historical data, trends, topics and so on.

All the gathered data then undergo the procedure of extracting only relevant information. Techniques as entity recognition or clustering takes place. Processed data is used as an input for the language models, such as GPTs, to generate response on the user's question. Language model will generate natural language text that match the user's inquire.

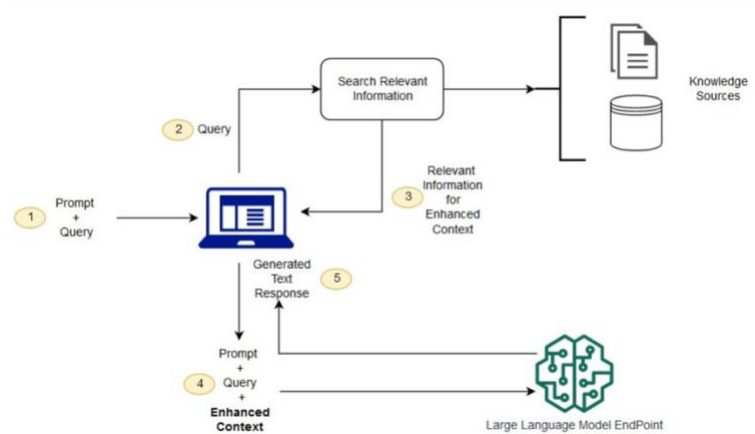


Figure 8 conceptual flow of using RAG with LLMs (source: aws.amazon.com.)

2.7 Importance of Custom ChatGPT

Custom versions of ChatGPT offer significant advantages and have grown increasingly important since their release, impacting various aspects of their use.

Innovation and Experimentation: One of the key benefits of custom ChatGPT is fostering innovation and experimentation. Users can design and develop new applications tailored to their specific needs. This not only boosts creativity but also allows users to explore AI capabilities beyond the already created pre-built models (Gaurav Gand, 2024).

Control Over Content: Having a GPTs enables users to ensure the accuracy of the generated content and to have control over it. This capability is crucial for correcting errors and reducing the risks of generating misinformation or experiencing AI "hallucinations" (Gaurav Gand, 2024).

Corporate Applications: GPTs also elevate AI applications within corporate settings. Companies can deploy personalized models for various functions such as customer service, creating technical documentation and reports, or driving personalized marketing campaigns (OpenAI, 2023).

3 Practical Part

This section shows the implementation and assessment of two different custom versions of ChatGPT. The first application is designed to assist users in creating CVs and resumes, helping individuals create their professional profiles without requiring advanced skills. The second application focuses on generating quizzes for students and teachers to prepare for the Japanese proficiency language tests from level N5 (equivalent to A1) to N3 (equivalent to B1). The performance and utility of these GPTs will be evaluated by specialists in the required fields. Specifically, the resume-oriented GPTs will be reviewed by HR specialists, while language preparation will be checked by the teachers in language schools and people that successfully passed the Japanese language test. The evaluation criteria will include specific metrics, which will be described. Additionally, evaluators will have the opportunity to contribute insights that they consider important to improve the functionality and effectiveness of the GPTs. Feedback will be provided from the candidates through email or direct conversations.

3.1 Resume and Cover Letter specific GPTs

Writing an effective CV and cover letter is crucial and should not be underestimated. An impressive resume does more than just assist job seekers in securing their ideal position faster; it also helps human resources personnel in assessing and selecting the most suitable candidate for the job. However, it can be challenging for applicants to create an engaging resume that showcases their employment history, skills, educational background, volunteer experiences, and other key information. It's important to explain these points clearly without making them too complicated, while keeping the original message intact.

The problem users may have while using basic ChatGPT is a lack of personalized content suggestions, and may not adapt well to users outside of their professional fields, GPTs can generate content but may lack the nuanced understanding needed for effective resume and cover letter writing.

3.1.1 Definition of the scope of the GPTs capabilities

The process of making customized ChatGPT does not require special skills or professional background for the users.

From the very beginning of starting of creating customized ChatGPT, the question of the purpose of the GPTs will be given. At that point, the prompt below has been provided to the model.

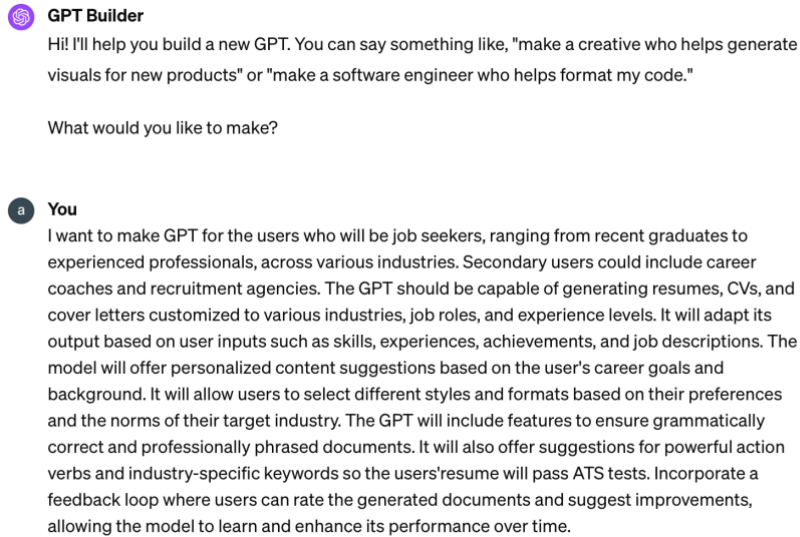


Figure 9 definition for resume GPTs (source: author)

The next steps will be defining the tone and language usage withing GPTs. Here creator needs to define, how it will communicate with users.

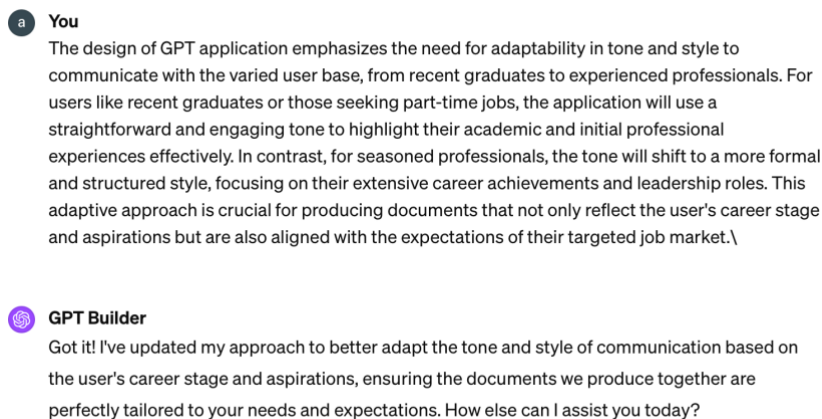


Figure 10 definition of tone in resume GPTs (source: author)

To make GPTs more user-friendly, it was instructed to adapt to the user experience in job hunting. While generating documents for juniors or students, then it is important to ask for more details about the university or personal background. When talking with skilled experts it would talk in professional language and ask more about their experience in the field.

3.1.2 Data collection and configuration

To ensure the GPTs remains relevant with the newest job market trends and best practices, it's essential to continually update its training data beyond the initial cut-off in April 2023. New documents, including advice, tips, and practical examples (resumes, CVs, and cover letters), have been sourced from leading employment platforms and publications such as LinkedIn, Indeed, Forbes, and similar. These sources are considered to be one of the best platforms not only for job hunting but

also for seeking useful recommendations and advice, including a wide range of resume and cover letter examples and templates.

Firstly, datasets for this GPTs include specific templates, which it should orient on. The main focus of the generated content should follow the structure of the given templates, specifically, separating lines for each part, bullet points, numbers, and percent into the achievement part. An example of one of the templates is II.

Secondly, beyond structural accuracy, the model is crafted to incorporate a human touch within the documents it generates. It uses advanced natural language processing techniques to compose cover letters that take personal achievements and career aspirations, making them compelling to human readers like potential employers or recruiters.

Before training, these documents underwent cleaning and pre-processing to remove irrelevant information, for example, personal contact details, so the GPTs would not be distracted by it. This process makes sure that the GPTs advice is up-to-date and relevant, offering useful tips and help to a wide range of people looking for jobs today.

This methodical approach to data enhancement and cleaning is crucial for maintaining the utility and relevance of the GPT as a key resource for individuals preparing job application documents in line with the latest market insights and practices.

3.1.3 Fine-tuning

For the fine-tuning process, GPTs offer automatic tools, directed to improve different fields, during the creation of the model.

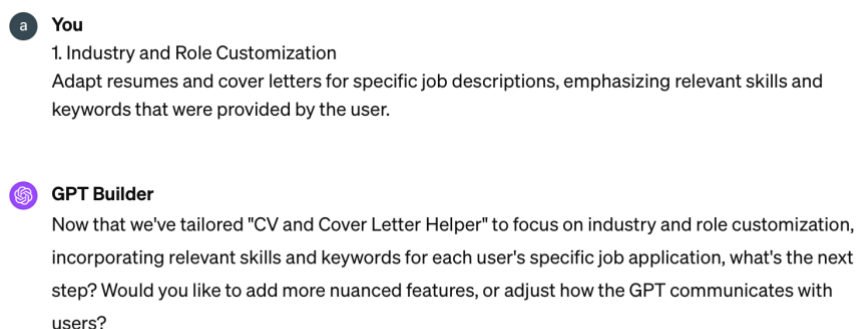


Figure 11 Fine-tuning GPTs for resume (source: author)

The first fine-tuning instruction added to the model, will adapt the resume based on the input given by the user, and make sure to include all the relevant information with the ability to expand it to make the output look more professional.

In the next steps, eight more conditions were added to the model, including formatting, optimal language, and tone.

To help users save time while using the model, the rule for asking for as much information of the user's background as possible was added. It helps junior or

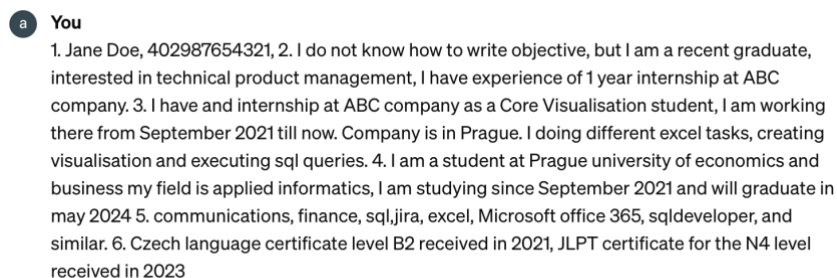
student-level users to be able to make the process of creating their resume easier, as they are more likely to have not done it before.

3.1.4 Testing of the GPTs

In this part, the testing of the model will be executed. The aim is to understand if the model will work correctly, according to all the instructions and data, that were given to it.

Firstly, the creation of a resume will be tested on the model. The aim of the GPTs will be to create a resume with a simple and structured format. As GPT-4 supports the creation and further downloading of Word documents, the resume will be made in this format.

The input below was introduced to the GPTs with basic information about the user, including education, previous work experience, skills, and languages.



The image shows a chat window with a single message from the user. The message is a list of six numbered items providing personal and professional details for a resume. The items are: 1. Name and ID (Jane Doe, 402987654321), 2. Statement of lack of objective writing experience but recent graduation and interest in technical product management with 1 year of internship at ABC, 3. Internship at ABC as a Core Visualisation student working on Excel tasks and SQL queries since September 2021, 4. Current student at Prague University of Economics and Business in Applied Informatics, starting in September 2021 and graduating in May 2024, 5. Skills in communications, finance, SQL, Jira, Excel, Microsoft Office 365, and SQL development, and 6. Language certificates: Czech B2 (2021) and JLPT N4 (2023).

a You

1. Jane Doe, 402987654321, 2. I do not know how to write objective, but I am a recent graduate, interested in technical product management, I have experience of 1 year internship at ABC company. 3. I have and internship at ABC company as a Core Visualisation student, I am working there from September 2021 till now. Company is in Prague. I doing different excel tasks, creating visualisation and executing sql queries. 4. I am a student at Prague university of economics and business my field is applied informatics, I am studying since September 2021 and will graduate in may 2024 5. communications, finance, sql,jira, excel, Microsoft office 365, sqldeveloper, and similar. 6. Czech language certificate level B2 received in 2021, JLPT certificate for the N4 level received in 2023

Figure 12 Testing GPTs for resume (source: author)

Based on this information, the model generated the output that can be seen in the appendices I. This output contains all the needed information and is generated along with the required formatting rules.

For now, generated output effectively combines formatting, needed information inclusion, and logical structuring. However, the result should still be edited by the user manually for small changes, even though GPTs can generate documents in a “.docx” format, it is still not well integrated in the Microsoft Word formatting techniques.

The testing part shows that resume and cover letter-specific GPTs work well for the purpose it was created. All the answers have a strict structure with headers, bullet points, and well-organized paragraphs. Information in the outputs is up-to-date and responds to recent news and articles from the professional world. Moreover, the model asks for additional information from the user to make the output to be as relevant as possible.

3.1.5 Criteria for the further evaluation

The criteria, for evaluating the model's performance by HR specialists, were chosen with a focus on its ability to create specific resumes and cover letters. Inspired by templates seen online and insights gathered from discussions with HR

professionals, these criteria were formulated to show essential elements required for effective job application documents. While existing resources lacked specific guidelines for GPTs, this table represents a collective combination of industry standards and tailored expectations for the functionality of this AI tool in resume and cover letter generation.

The table below shows the criteria name, description, and score with the scale from one to ten, where one is the lowest possible ranking, and ten is the highest possible.

Criteria	Description	Score
Relevance	Generated resume, CV or cover letter is relevant to the information, skill, experience, and qualifications provided by the user.	1-10
Formatting	Output is visually appealing, with clear section headings and bullet points.	1-10
Human Touch	Output captures the candidate's personality and unique selling points. It is hard to understand if it was written by the AI or human.	1-10
Accuracy	Details such as dates, job titles, and educational qualifications are presented accurately.	1-10
Creativity	Output demonstrates creativity and originality in presenting qualifications and achievements.	1-10
Coherence	Text is well-organized, easy to read, and does not contain grammatical errors.	1-10
Keywords usage	Output includes relevant keywords and phrases for the target position.	1-10
Customization	Generated output is tailored to match the individual's unique experiences and qualifications.	1-10
Conciseness	Output is concise and avoids unnecessary verbosity or repetition.	1-10

Table 1 Evaluation table for resume GPTs (source: author)

3.1.6 Evaluation of the resume specific GPTs

The evaluation reviews were collected from workers in the human resource field. Candidates were chosen based on their job background in the required role. They have experience in employee engagements, early talent engagement programs, managing employee life cycles and employee relations and management.

To make the evaluation process more diverse, people, who were chosen for the evaluation, have working in the industry for different amounts of time from 3 years to 6 years of professional experience.

Each candidate evaluated the model by himself, including writing prompts, creating his own resume, and asking relevant questions. However, to make the evaluation process more appealing each candidate was provided with two CVs and two cover letters. With the information, that first document was written by a human. While the second document was written and generated completely by artificial intelligence, based on the information received from the user.

The first tester is a candidate with more than five years of experience in the recruiting field. Based on the LLM's performance, the following scores were stated.

Criteria	Description	Score
Relevance	Generated resume, CV or cover letter is relevant to the information, skill, experience, and qualifications provided by the user.	8
Formatting	Output is visually appealing, with clear section headings and bullet points.	9
Human Touch	Output captures the candidate's personality and unique selling points. It is hard to understand if it was written by the AI or human.	5
Accuracy	Details such as dates, job titles, and educational qualifications are presented accurately.	7
Creativity	Output demonstrates creativity and originality in presenting qualifications and achievements.	7
Coherence	Text is well-organized, easy to read, and does not contain grammatical errors.	8
Keywords usage	Output includes relevant keywords and phrases for the target position.	7
Customization	Generated output is tailored to match the individual's unique experiences and qualifications.	9
Conciseness	Output is concise and avoids unnecessary verbosity or repetition.	5

Table 2 Evaluation of resume GPTs from the first candidate (source: author)

As reflected in the table above, the custom version of ChatGPT received an average score of 7 points, with its overall performance evaluated as good. The tester highlighted several advantages of the custom ChatGPT.

First, there is the significant time savings for individuals who cannot devote much time to crafting a resume. The GPTs operates efficiently, generating high-quality outputs without needing further prompts to enhance previous responses.

Additionally, the candidate noted that, compared to the free version of ChatGPT, the custom version better understands user needs. For example, when asked to create a resume and suggest additional skills relevant to the user's job position, this version accurately incorporated both technical and soft skills based on the information provided about the user and the job advertisement. The creativity of the GPTs was also praised, especially in the creation of cover letters that met all specified requirements and exhibited a human-like quality.

However, there were some disadvantages noted as well. Mainly it was the occasional repetition of certain words or phrases, which was seen as discouraging because it made the resumes appear less professional. Although the cover letter was considered very human-like, the resumes were judged more critically. It was noticeable in some outputs that the resumes were crafted by artificial intelligence, which detracted from their overall quality.

In terms of the comparison between the advanced preparation of the cover letter and resume: for the cover letter, the candidate found it challenging to understand which was written by the AI and eventually chose based on intuition. This choice was correct, and the candidate successfully identified the cover letter generated by the custom ChatGPT. As for the resume, the candidate was able to recognize the one produced by the AI. In the pictures provided, the candidate highlighted words and phrases that, in their view, indicated signs of AI involvement.

Motivated Project Manager with 9 years of experience in diverse aspects of project management, product leadership, and business development. Seeking to leverage **my expertise** in client relations, marketing strategy, and project execution to drive success and achieve goals in a dynamic environment. **Adept at** navigating complex projects with meticulous attention to detail and a proven record of delivering quality, profitable projects on time and within budget.

Figure 13 Resume evaluation by the first candidate (source: author)

Thank you for considering my application for the Senior Project Manager role. With over 10 years of professional experience in leading significant projects across complex strategy development, international event management, and high-stakes negotiations, I am **enthusiastic about the opportunity** to contribute to your team's success. My background aligns perfectly with the requirements for this role, and **I am eager** to leverage my skills to drive innovative outcomes at your company.

Figure 14 Cover Letter evaluation by the first candidate (source: author)

According to the expert, the main indicators that suggest a document was generated by artificial intelligence include the use of clichés such as "eager" and "enthusiastic," and highly repetitive use of the words "I," "me," and "my." However, it was noted that if the observer were not specifically analyzing and comparing these documents for the thesis, and instead received the resume or cover letter as a response to a job advertisement in real life, it would be difficult to understand or notice that the document was automatically generated.

These insights suggest that while the custom ChatGPT offers benefits in terms of time efficiency, accuracy, and creativity, attention to reducing repetitive content and enhancing the naturalness of the resume output could further improve its utility.

The next candidate has three years of experience in the field of human resources and coordination. Based on the evaluation and given criteria, the next scores were estimated.

Criteria	Description	Score
Relevance	Generated resume, CV or cover letter is relevant to the information, skill, experience, and qualifications provided by the user	8
Formatting	Output is visually appealing, with clear section headings and bullet points	7
Human Touch	Output captures the candidate's personality and unique selling points. It is hard to understand if it was written by the AI or human.	6
Accuracy	Details such as dates, job titles, and educational qualifications are presented accurately.	8
Creativity	Output demonstrates creativity and originality in presenting qualifications and achievements.	6
Coherence	Text is well-organized, easy to read, and does not contain grammatical errors.	9
Keywords usage	Output includes relevant keywords and phrases for the target position.	8
Customization	Generated output is tailored to match the individual's unique experiences and qualifications.	8
Conciseness	Output is concise and avoids unnecessary verbosity or repetition.	6

Table 3 Evaluation of resume GPTs from the second candidate (source: author)

As can be seen in the table above, scores have almost the same value, as the previous candidate.

The average score here is also 7 and the overall performance of the model was said to be excellent. In this evaluation process candidate mainly aimed to understand the time-saving aspect of this custom version of ChatGPT.

Firstly, talking about advantages, formatting, and overall organization of the resume and cover letters were mentioned. Required structure was observed, therefore there was no need to fix output manually. Which was mentioned to be a positive point for time-saving.

Secondly, the next positive part was the accuracy and relevance of the generated output. For the given background of the user and needed job description, the custom version was able to generate clear and accordant answer.

Between disadvantages, the problem of human touch took place. For the candidate, sometimes it was time-consuming to fix or rewrite words and phrases, that sounded too scientific or were too difficult for the resume, which means, people do not usually use this type of lexicon.

The next disadvantage was told to be a lack of creativity in the field of creating resumes. For cover letters, GPTs was able to create unique paper, with all the needed information. For the CV it sometimes used lots of cliché words and phrases, which needed to be either fixed or removed. This would also affect the time-saving aspect because the user would spend time scanning and correcting given output.

Same as the previous candidate, this tester was told to evaluate two resumes and two cover letters to identify which one was created by human and which one by artificial intelligence.

In the picture below it can be seen that the candidate has successfully defined resume generated by the GPTs. The keywords and phrases, that show the touch of artificial intelligence, were also highlighted.

BigWood LLC (PhosAgro OJSC), Kirovsk, Russia
Ski Resort Brand Manager,
Jul 2012 – Mar 2016

- Boosted the resort's global web presence by 40% with aggressive marketing.
- Developed and executed annual marketing plans, aligning with strategic business objectives.
- Enhanced local operations through innovative marketing approaches, significantly increasing brand awareness.
- Negotiated new contracts with stakeholders, saving up to \$20,000 for the annual budget.

Figure 15 Resume evaluation by the second candidate (source: author)

• JIRA	• Unreal Engine 4	• OS/iOS
• Google Analytics	• Image Production	• Keynote / iWork / Pages
• PowerPoint	• Websites	• Word
• Microsoft Office 365	• POS / Videos	• Excel
• Adobe Photoshop	• Email Marketing	• PowerPoint

Figure 16 Resume evaluation by the second candidate (source: author)

The candidate noted that while the overall structure of the generated document was good with only a few minor mistakes, the "Technical Skills" section was quite repetitive. Specifically, he pointed out that the resume lists Microsoft Office 365, which already includes PowerPoint, Excel, and Word, but these skills were redundantly listed individually in separate points.

The expert successfully identified the resume as automatically generated. However, the task of distinguishing the cover letter was not completed accurately. The phrases initially thought to be produced by the GPTs were part of the cover letter written by a human. Nevertheless, the candidate still highlighted some words and phrases in the document that were automatically generated, which can be seen in the picture below.

Thank you for considering my application for the Senior Project Manager role. With over 10 years of professional experience in leading significant projects across complex strategy development, international event management, and high-stakes negotiations, I am enthusiastic about the opportunity to contribute to your team's success. My background aligns perfectly with the requirements for this role, and I am eager to leverage my skills to drive innovative outcomes at your company.

In my career, I've worked in various roles that match what you're looking for. As a Brand Manager, collaborating with cross-functional teams, I managed and optimized key communication channels and spearheaded strategic marketing campaigns across digital platforms. These initiatives resulted in a notable increase in brand visibility, propelling website pageviews by 40% and achieving a remarkable 300% surge in new customer acquisition within a single fiscal year.

Figure 17 Cover Letter evaluation by the second candidate (source: author)

3.1.7 Reasons to use custom version of ChatGPT for generating resume

Candidates who were evaluating the performance of customized language models were also asked to explain why they would prefer the custom version to the standard one and if they noticed any differences between them. Since their answers were quite similar, responses were combined into one comprehensive list. This helped to understand the common advantages of using customized models according to the candidates.

1. **Specialized Knowledge:** It's difficult to identify exactly what data the usual ChatGPT version was trained on, which generally leads to broad answers, which cover many topics, including human resources knowledge. However, it often lacks depth and may not generate the desired answer on the first attempt. This suggests that a custom version of GPT could be more effective for producing specific outputs, as it would be trained on well-defined datasets known to both the developer and the users.
2. **Time-Saving Aspect:** The custom version is designed to generate a specific type of resume—simple and minimalistic, containing only essential information without any photos or special design elements. Candidates noted that this feature could be particularly efficient for users who need to create multiple resumes daily. Manual writing can be time-consuming and exhausting, especially when applying to numerous job positions. Using this custom ChatGPT provides a quicker and less complicated process, as it generates the required output with a single prompt, delivering the resumes in the desired format with all necessary information included.
3. **Integration and Assistance for Beginners:** While GPT-4 can scan and correct documents, this custom version excels in its ability to integrate with provided documents and convert them into the correct format. This feature is particularly beneficial for beginners in the job application process. The custom version offers guidance by initially asking users several questions about their background. This customized approach not only generates the output quickly but also simplifies the process for students or junior specialists, making it easier for them to produce professional documents efficiently.

In conclusion, GPTs can be beneficial for beginners in the job-seeking world. While other versions of ChatGPT have their uses, customized model that helps users quickly create their portfolios can save plenty of time. For advanced and skilled professionals, this can primarily assist by analyzing documents, adding new details, correcting formats, and performing other necessary tasks efficiently. This tailored approach ensures that all users, regardless of their expertise level, can benefit from the model's capabilities.

3.2 JLPT Preparation Helper GPTs

Official Japanese language proficiency test has always been considered a challenge, even for the lowest level possible. A large amount of various Chinese characters, known as kanji, vocabulary, reading, and grammar, are the parts that a person should know perfectly to be able to pass the exam. However, studying language is tough not only for students but also for teachers. The need to create different tests, quizzes or assessments could be a draining and time-consuming process. Despite the fact, that there are many preparation materials and books, it can be hard to structure everything in one place and find many different test examples for practice.

The purpose of this LLM is to assist people with making quizzes and tests in preparation for the official exam. The model will offer users several questions that they will answer, and in the end, the results and mistakes will be evaluated. Users can generate tests, and quizzes, ask for explanations of the grammar rules, or generate a list of needed vocabularies or topics for different levels.

3.2.1 Definition of the scope of the GPTs capabilities

For this version, the first step will be defining the goal. As can be seen below, the goal is preparation for the official language test.

a You
GPT for preparation for JLPT test. I want it to make tests that will look like the real-exam one. And generate different quizzes for users. Chat could be used by both, student and teachers.

Figure 18 JLPT GPTs creation (source: author)

After that, the question of language tone for communication was asked. Here the style of interaction should be user-friendly because the custom version will be headed to help users who may have zero or beginner knowledge.

a You
Tone should be user-friendly, as user could be a complete newer to the Japanese language. Explanations of the answers and question should be as clear as possible, with translation and kanji reading.

Figure 19 JLPT GPTs creation, 2 step (source: author)

For this GPTs, the question of approach for generating quizzes and tests should be solved. Here it has been decided to always ask users about their preferences in quiz and test structure and content, to make sure output will be generated along with users' needs.

3.2.2 Fine-tuning of the GPTs

In this part, several aspects of the custom ChatGPT were fine-tuned to make its performance more accurate.

Firstly, content focus. This LLM should be able to generate content for all the parts of the real exam, which includes, grammar, vocabulary, reading comprehension, and kanji, except the listening comprehension as audio generation is not allowed.

Secondly, the format of the questions. During real exams, the English language is not allowed, all the papers should be written in the Japanese language. That is why, it was fine-tuned to generate tests for JLPT only in the Japanese language, to make its performance and preparation help more advanced.

Thirdly, response details. As for beginners as for the more skilled users, GPTs was fine-tuned to provide output with a proper format. For example, for the tests, it will be, the test itself, the explanation, which will include the answers for the test, and the feedback for the user, which will show all the mistakes and its review for further studying.

3.2.3 Data collection and pre-processing

The important step in developing a comprehensive language model for the Official Japanese Language Proficiency Test preparation involves extensive data collection gathering. This phase is important for getting a diverse and rich dataset that cover a wide array of language skills assessed in the JLPT, including vocabulary, grammar, kanji, and reading comprehension. The key sources for this model include:

- Educational materials – to improve the accuracy and variety of the GPTs, the author included specific grammar rules and definitions for different levels of Japanese proficiency. This addition helps teachers create focused quizzes and enables self-learners to better understand and apply Japanese grammar. This feature was added because earlier models like GPT-3.5 and GPT-4 were sometimes inaccurate in presenting grammar, providing incomplete information, or generating content at the wrong difficulty level.
- Past year exam papers – creating datasets of past year’s JLPT exam papers allows to understand the exam structure and format, question types, and defining difficulty for each level of exam. Examples were provided by the Japan Foundation and Japan Educational Exchanges and Services. Datasets include past year exam papers available through official JLPT websites and additional exams sourced from internet forums where users have shared scanned copies. This collection strategy provides a broad spectrum of exam formats and question types, facilitating a deeper understanding of exam complexity across different years and levels. Training on these diverse sources enables the generation of practice materials that closely mimic actual exam scenarios, providing learners with realistic and varied test preparation.
- Vocabulary and Kanji Lists – gathering and creating vocabulary and kanji lists, that were found on different websites¹. It includes documents with words, their reading, and writing patterns. These lists were personally compiled using a variety of online resources to close the gap where official materials are lacking, especially in providing exhaustive and level-specific vocabulary and kanji words. By creating and refining these lists, the GPT is

¹ For the purposes of this thesis author used websites: jlptsensei.com, jlptstudy.net, tanos.co.uk/jlpt/, jlpt.jp

trained to focus precisely on the requirements of each JLPT level, ensuring that the content generated is both accurate and highly relevant to exam preparation. This categorization helps in tailoring the model to produce outputs that are directly aligned with the learning objectives of the JLPT syllabus.

After data collection, the stage of pre-processing took place. In this part, the datasets were ensured to have high quality and consistency for better model performance.

The key steps include:

- Text normalization, involving removing all the unnecessary or irrelevant information, for example, footnotes and links, that are not directly related to the JLPT content.
- Data labeling and categorization include using the uniform format for labels (Level, JLPT, name of the dataset) and defining the content type.

3.2.4 Testing of the GPTs

In this section, the language model will be tested to ensure it works according to defined knowledge.

Firstly, the quiz generation will be tested. This part was generated by the GPTs as was expected. It provided three different types of questions along with four options.

Question 1: What does the word "あさ (asa)" mean in English?
A) Evening
B) Night
C) Morning
D) Afternoon

Question 2: Which of the following means "book" in Japanese?
A) ほん (hon)
B) いぬ (inu)
C) ねこ (neko)
D) くるま (kuruma)

Question 3: The Japanese word "きょうだい (kyoudai)" translates to which of the following in English?
A) Cousin
B) Sibling
C) Uncle
D) Friend

Figure 20 Testing quizzes in JLPT GPTs (source: author)

Moreover, in the beginning model was trained to always provide feedback based on the users' answers, including overview of all the responses and correction and explanation of the incorrect ones. This feedback was also provided by the model at the end of the quiz.

The next testing part will be generating the JLPT test example for the basic level (N5) in the grammar part.

Question 1: この本は面白いです _____、少し難しいです。

- A) けれども
- B) と
- C) から
- D) ので

Question 2: 私は魚が _____が、肉はあまり好きじゃないです。

- A) 好きです
- B) 好きじゃないです
- C) 食べます
- D) 食べません

Question 3: 明日のパーティーにはケーキ _____ピザを持っています。

- A) や
- B) と
- C) を
- D) でも

Figure 21 Testing exam test in JLPT GPTs (source: author)

This part was also generated as expected. Questions are grammatically correct, and the structure goes along with the real test.

The last part that will be tested is the ability to provide correct information and explanations for theoretical questions.

The model was asked to provide two grammatical rules for the basic level of language test with an explanation.

1. Using が (ga) with 好き (suki) and 嫌い (kirai)

Rule Explanation:

In Japanese, when expressing likes (好き suki) or dislikes (嫌い kirai), the particle が (ga) is used to mark the object of affection or dislike, rather than the often used を (wo) particle that marks the direct object of a verb. This is because 好き (suki) and 嫌い (kirai) are na-adjectives, not verbs, so the thing you like or dislike is treated as the subject of the sentence rather than the object.

Example:

- 猫が好きです。(Neko ga suki desu.) - I like cats.
- 辛い食べ物が嫌いです。(Karai tabemono ga kirai desu.) - I dislike spicy food.

2. Using て-form to connect sentences

Rule Explanation:

The て-form (te-form) of verbs can be used to connect multiple actions in the same sentence, indicating a sequence of actions or states. This form is incredibly versatile and is a foundational aspect of Japanese grammar. It allows for connecting verbs without necessarily implying causation or reason, simply stating that actions occur in sequence.

Example:

- 朝ご飯を食べて、学校に行きます。(Asagohan o tabete, gakkou ni ikimasu.) - I eat breakfast and then go to school.
- In this sentence, the て-form of 食べる (taberu, to eat) is used to connect eating breakfast with going to school, indicating that these actions happen one after the other.

Figure 22 Testing basic knowledges in JLPT GPTs (source: author)

As can be seen, information was provided as needed. Grammatical rules are on the basic level, and detailed explanations with examples were generated correctly.

In conclusion, the tested part showed that the GPTs work right, with no grammatical or factual mistakes. Generated answers were accurate and well-structured. All the

rules, including feedback, mistakes overview, and formatting, were verified. Now, the model is ready to be tested by the experts.

3.2.5 Criteria for the further evaluation

The criteria below were estimated based on the main purposes that the model should follow. Criteria for this GPTs were mainly made based on the author's experience of preparing for language tests and with the help of some experts, as, in advance, they were asked why they would need to use GPTs instead of general ones. The aim is to define whether this tool can be used for studying purposes without worsening users' knowledge of the language and give them an opportunity to effectively prepare for the test.

Criteria	Description	Score
Grammar and spelling	Grammatical and spelling rules are fulfilled. Sentences, words, and lists are structured as needed, and don't contain mistakes or other issues.	1-10
Accuracy	The generated answer meets all requirements and contains all the necessary information, aligning perfectly with the specified measures.	1-10
Formatting and structure	The generated answer is well-structured and formatted, presenting information in a clear and organized manner. Each point flows logically from the previous one, ensuring readability.	1-10
Time-saving	The usage can be evaluated as a time-saving, by generating answers quickly and correctly.	1-10
Variety of Question Types	Generated answers are diverse, quizzes and tests contain different types of questions allowing to practice more efficiently.	1-10
Alignment with JLPT	Generated tests have the same structure and types of questions as a real-life language exam.	1-10
Adaptability and Customization	Flexibility of the model, ability to adapt to the user's needs and preferences, for example, focusing on certain grammar points or vocabulary themes.	1-10

Table 4 Criteria for JLPT GPTs evaluation (source: author)

3.2.6 Evaluation of the JLPT Helper GPTs

For the evaluation of a JLPT preparation-focused language model, candidates with at least a JLPT N2 certificate, equivalent to a B2 level of language proficiency, were selected.

One of the candidates is an experienced Japanese language teacher with nearly 15 years of experience in teaching language at university. She runs her language school that serves more than sixty students. Her responsibilities also include preparing students for the JLPT exams. Extensive work and study experience in Japan demonstrate her fluency and expertise in the language.

Other selected candidates have backgrounds in living and studying in Japan, with academic credentials in Japanese history and culture. One of these candidates graduated from Doshisha University and has certifications for all levels of the JLPT exams. Another is currently residing and working in Japan, having previously taught the Japanese language.

During the evaluation phase, candidates were allowed to independently test the custom ChatGPT based on the previously established criteria. The testing involved creating quizzes and tests for JLPT, assessing grammar, measuring creativity, and comparing the advantages of this specialized version to the standard, freely available ChatGPT versions.

The first candidate to provide an evaluation is a teacher. The testing made by this candidate included creating quizzes, designing tests, and assessing grammar correctness. Below, a table of scores is provided to summarize the results.

Criteria	Description	Score
Grammar and spelling	Grammatical and spelling rules are fulfilled. Sentences, words, and lists are structured as needed, and don't contain mistakes or other issues.	8
Accuracy	The generated answer meets all requirements and contains all the necessary information, aligning perfectly with the specified measures.	9
Formatting and structure	The generated answer is well-structured and formatted, presenting information in a clear and organized manner. Each point flows logically from the previous one, ensuring readability.	9
Time-saving	The usage can be evaluated as a time-saving, by generating answers quickly and correctly.	8
Variety of Question Types	Generated answers are diverse, quizzes and tests contain different types of questions allowing to practice more efficiently.	7

Alignment with JLPT	Generated tests have the same structure and types of questions as a real-life language exam.	8
Adaptability and Customization	Flexibility of the model, ability to adapt to the user's needs and preferences, for example, focusing on certain grammar points or vocabulary themes.	8

Table 5 Evaluation of JLPT GPTs from the first candidate (source: author)

Based on the evaluation table, the custom ChatGPT version received an average score of 8. Positive feedback about the GPTs highlighted several strengths.

Answers were generated with high accuracy in grammar and spelling, showing almost no errors in word choice or sentence structure. A key benefit of this accuracy is that it eliminates the need for extensive double-checking of the generated output. The GPTs was capable of generating a wide range of questions for quizzes and tests, without any repetition or irrelevant content. It demonstrated significant creativity, consistently producing engaging and varied tasks.

Another notable advantage was its ability to closely recreate the format of actual JLPT exams, which do not include any English language. This aspect was particularly valued as it helps in creating practice materials that accurately reflect the structure and style of the exams, thus being better for preparing candidates for the real test conditions.

The only disadvantage noted in the feedback was occasional confusion between the JLPT levels, particularly between the beginner level N5 and the slightly more advanced N4 level. However, the candidate pointed out that this issue could benefit students by exposing them to more challenging questions. This exposure might better prepare them for the tougher questions that could potentially appear in the actual exams, therefore being a hidden advantage in their exam preparation.

The next candidate is a person who has the experience of self-preparation for all levels of JLPT exams. With the successful passing of all the exams. With one year of experience in studying in a Japanese university. The table below shows the scores estimated.

Criteria	Description	Score
Grammar and spelling	Grammatical and spelling rules are fulfilled. Sentences, words, and lists are structured as needed, and don't contain mistakes or other issues.	9
Accuracy	The generated answer meets all requirements and contains all the necessary information, aligning perfectly with the specified measures.	10
Formatting and structure	The generated answer is well-structured and formatted, presenting information in a clear	9

	and organized manner. Each point flows logically from the previous one, ensuring readability.	
Time-saving	The usage can be evaluated as a time-saving, by generating answers quickly and correctly.	7
Variety of Question Types	Generated answers are diverse, quizzes and tests contain different types of questions allowing to practice more efficiently.	7
Alignment with JLPT	Generated tests have the same structure and types of questions as a real-life language exam.	8
Adaptability and Customization	Flexibility of the model, ability to adapt to the user's needs and preferences, for example, focusing on certain grammar points or vocabulary themes.	7

Table 6 Evaluation of JLPT GPTs from the second candidate (source: author)

In this review candidate has mentioned several advantages, that, in his opinion, play huge role in preparation to the language exam.

Firstly, a significant advantage of the custom ChatGPT is its ability to save time and provide immediate reviews. As previously mentioned, exam preparation can be tough due to the vast number of websites and papers a student must go through to solve various questions. While now, there is no need to spend hours surfing the web. Students can simply specify their questions and the themes they need to explore. This tool is also incredibly useful for conducting quick reviews right before tests or for summarizing and revisiting studied materials.

Secondly, the candidate was particularly impressed by the custom version's ability to provide mandatory feedback on every question answered, without needing specific prompts or instructions to do so. This feature helps students better understand their mistakes and learn the appropriate topics to improve their language skills. This feedback mechanism ensures that learning is not only about answering questions but also about understanding and correcting mistakes, which is crucial for effective language learning.

Alongside the benefits, the candidate also mentioned that despite the tool's effectiveness in preparation and review, it is essential for students to engage in face-to-face communication with a real teacher or an expert to ensure high-quality exam preparation. The limitation noted here is that users cannot rely on artificial intelligence, even though it can be a significant help in enhancing knowledge. This highlights the importance of a balanced approach, combining the efficiency of AI tools with the personalized guidance and expertise of human instructors to achieve the best educational outcomes.

The last candidate's evaluation can be seen in the table below. This person currently lives in Japan, even though he is not working in the education field, he has a background of teaching and preparing people for the basic levels of the Japanese language.

Criteria	Description	Score
Grammar and spelling	Grammatical and spelling rules are fulfilled. Sentences, words, and lists are structured as needed, and don't contain mistakes or other issues.	8
Accuracy	The generated answer meets all requirements and contains all the necessary information, aligning perfectly with the specified measures.	10
Formatting and structure	The generated answer is well-structured and formatted, presenting information in a clear and organized manner. Each point flows logically from the previous one, ensuring readability.	6
Time-saving	The usage can be evaluated as a time-saving, by generating answers quickly and correctly.	6
Variety of Question Types	Generated answers are diverse, quizzes and tests contain different types of questions allowing to practice more efficiently.	7
Alignment with JLPT	Generated tests have the same structure and types of questions as a real-life language exam.	7
Adaptability and Customization	Flexibility of the model, ability to adapt to the user's needs and preferences, for example, focusing on certain grammar points or vocabulary themes.	7

Table 7 Evaluation of JLPT GPTs from the third candidate (source: author)

The candidate gave the JLPT helper GPTs an average score of 8 and highlighted three primary advantages that significantly help in exam preparation.

First, review tool. The main advantage noted was the tool's effectiveness as a review mechanism, which reinforces knowledge of the studied topics. This is particularly beneficial in classroom settings where students might be asked to answer questions or complete quizzes on grammar and vocabulary topics. The ability to review material helps strengthen their understanding and preparation for exams.

Second, adaptability and question formatting. Another useful functionality is the adaptability of the tool and the format of questions generated by the AI. The candidate appreciated that the outputs were accurate and relevant to the prompts

given. If the candidate wanted to modify or add specific details to the questions, the custom version was able to accommodate these changes quickly and accurately. Third, immediate feedback. Similar to the previous candidate, the immediate feedback provided after each quiz or test was highly rated. This feature is beneficial for students as it helps them understand their mistakes clearly. For teachers, it saves time, a significant advantage since they often have multiple tasks and lessons to prepare.

Regarding the disadvantages, the candidate repeated previous concerns about depending too much on artificial intelligence. There is a concern that users might rely heavily on the content generated by AI, which could limit their ability to think critically and identify their areas of weakness. Despite this issue, the candidate suggested using the custom version of GPTs alongside traditional teaching methods provided by a language teacher. This combined approach, the candidate believes, would give the most effective learning results.

3.2.7 Reasons to use custom version of ChatGPT for preparation JLPT

Candidates were able to test both the standard and the custom version of ChatGPT, noting several key differences.

First, they observed that the custom version provided a structure and language format highly similar to the actual exam. While using the standard version of ChatGPT, candidates noted that the structure and language of the questions were often incorrect as they were in English, whereas the real exam is conducted only in Japanese. The custom version, however, could perfectly recreate the exam format without any need for users to specify this requirement. This feature is particularly beneficial as it offers a realistic preparation experience.

Second, the custom version is better in providing feedback. It consistently reviews the answers given by users, offering insights into potential grammar and vocabulary areas that need to be improved. This feedback is not only immediate but also educational, helping users enhance their knowledge. In contrast, the standard version of ChatGPT typically does not provide such feedback, making the custom version more suited for educational purposes.

Thirdly, for teachers, this custom version of ChatGPT allows the creation of more specific and individualized questions tailored to the student's language proficiency levels. Candidates with teaching experience mentioned that the standard version often failed to deliver specialized tests with the necessary structure. Meanwhile, the custom ChatGPT could analyze the given input and generate more detailed and professional responses, making it a valuable tool for educators.

In conclusion, the feedback from the candidates indicates that this custom version of ChatGPT is a valuable tool for preparing for language exams. It is created with detailed information about the JLPT exam, including a huge number of papers and tests from past years, and it offers users specialized knowledge to enhance their language learning progress. Furthermore, it is recommended to use this tool in conjunction with a teacher or additional materials. This approach allows learners to view problems and topics from various perspectives, which can further improve their understanding and proficiency in the language.

Conclusions

The goal of the thesis was to develop customized ChatGPT tailored to specific fields: job seeking, and preparation for the Japanese Language Proficiency Test, and critically evaluate the performance of these GPTs.

In the theoretical part, the concept of large language models and the mechanics of prompt generation were explored in depth. This provided a solid foundational understanding necessary for the practical application.

The practical part of the thesis involved creating two distinct custom versions of ChatGPT. The process began with defining the specific purposes of each GPTs. This was followed by data collection and pre-processing to ensure the models had relevant and high-quality data to learn from. Subsequent steps included fine-tuning the GPTs and conducting tests to confirm their functionality. Therefore, specific criteria for evaluation were established and clearly outlined.

For each customized ChatGPT, candidates and experts with relevant knowledge and experience were selected to assess the models. These individuals evaluated the models based on their practical utility. Their feedback was then meticulously documented and analyzed. The GPTs designed for resume and cover letter generation received positive feedback. While it did not significantly innovate beyond basic question-answering in the field, its ability to generate resumes and cover letters in the specific format was noted as particularly useful. It enabled the creation of simple, effective resumes with a human-like touch, requiring minimal effort from the user. The GPTs intended for JLPT preparation was highly praised by experts. It excelled in creating realistic test simulations and was recognized for its accuracy, notably its lack of grammatical errors commonly found in the standard versions of ChatGPT. Additionally, the structure and format of the tests and quizzes it generated were well-received for their comprehensiveness and correctness, reflecting the specific data provided.

Writing this thesis presented challenges, particularly in data collection and feedback gathering. For the resume-oriented GPTs, finding a large number of consistently formatted resumes and comprehensive cover letters was difficult. For the JLPT-focused GPTs, compiling extensive lists of tests, vocabulary, and kanji was problematic, especially with some resources restricted by copyright issues. Additionally, collecting feedback was challenging due to the busy schedules of experts and time zone differences complicating coordination for interviews and discussions.

In conclusion, the thesis has outlined the diverse applications of newly launched custom GPTs, offering valuable assistance in professional, educational, and creative fields. From resume generation to language exam preparation, this tool provides users with versatile support. Moreover, it has the potential for enhancing lesson and lecture preparation for educators. As the technologies evolve, their impact on daily tasks is expected to grow, fostering more efficient workflows in the future.

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Appendices

Appendix A: Data for GPTs training

Because of the large size and content of the data that was presented to the GPTs in the process of creating, Author decided to put everything in the GitHub repository, which can be found below.

<https://github.com/SashaMatveeva/GPTs-configure-files>

In this link, there are two folders: GPTs-for-JLPT, where all the data for the language GPTs are stored; GPTs-for-resume, where data for resume GPTs were added, there are also documents that were introduced to the experts for evaluation.

Appendix B: Resume Example

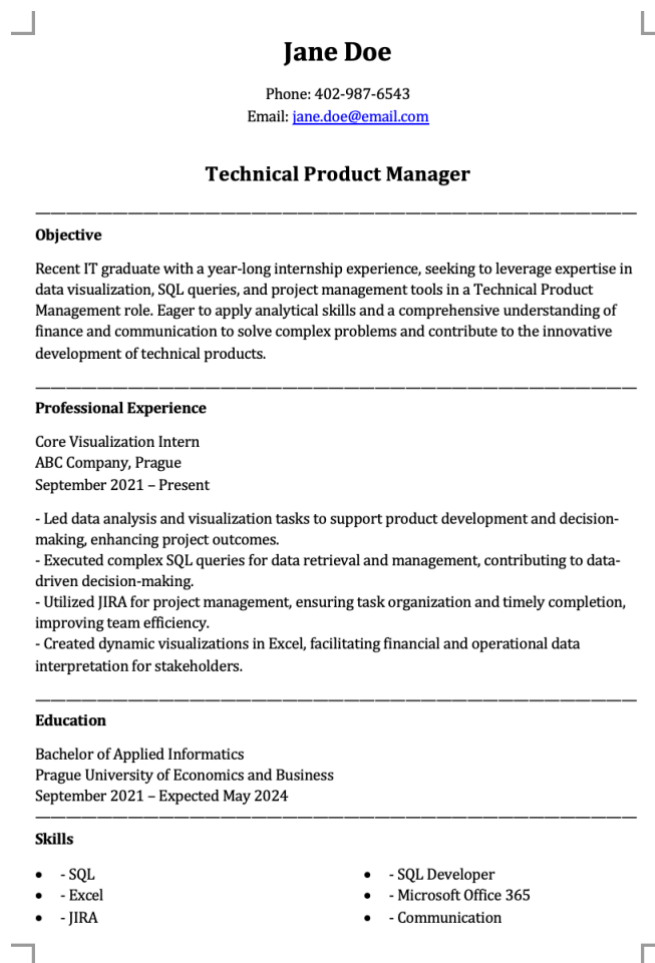


Figure 23 Generated resume (source: author)

Appendix C: Template example

Name Surname

PROFESSIONAL TITLE

Address | Phone Number | e-mail

Objective

To obtain a challenging data scientist position in a dynamic and innovative organization where I can use my technical and analytical skills.

Experience

FlueroGen | Data Scientist20XX – 20XX

- Increased customer retention by 20%
- Reduced fraudulent charges by 50%
- Improved product recommendations and increased sales
- Drove business decisions and reduced operational costs

Pantheros Labs | Junior Data Scientist20XX – 20XX

- Cleaned and preprocessed data
- Conducted statistical tests for decision support
- Developed machine learning models for prediction
- Collaborated on A/B tests, increased click-through

Education

Jasper University, MS Data Science20XX

Major: Data Science | Minor: Machine Learning

Figure 24 Resume Template example (source: LinkedIn.com)